

SANTAREM, Brazil

WMO: 822440

Lat: 2.422S

Lon: 54.793W

Elev: 198

StdP: 14.59

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	71.5	72.1	67.6	102.3	83.9	68.4	105.2	82.9	19.0	79.9	18.0	80.3	2.3	150	0.371

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	14.5	93.5	77.5	93.0	77.6	91.7	77.8	80.2	87.3	79.9	87.1	79.4	86.8	11.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.9	151.0	83.5	77.4	143.8	82.8	77.4	143.3	82.7	44.0	87.7	43.5	87.5	43.1	87.2	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.4	16.8	14.5	70.2	95.2	1.5	1.3	69.1	96.1	68.2	96.9	67.3	97.7	66.2	98.7		
(5)			68.6	82.1	1.6	2.1	67.4	83.6	66.5	84.8	65.5	85.9	64.4	87.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.1	81.6	80.8	80.8	80.8	80.8	80.9	82.4	84.0	84.0	84.4	84.0	83.3
	DBStd	2.47	2.75	2.35	2.14	1.88	1.67	1.49	1.48	1.54	1.68	2.11	2.15	2.60
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11707	980	863	955	925	956	925	959	1005	1019	1067	1021	1032
	CDD65	6232	515	443	490	475	491	475	494	540	569	602	571	567
	CDH74	66340	5197	4131	4549	4441	4631	4589	4874	6125	6991	7453	6893	6466
	CDH80	25876	1859	1278	1363	1323	1429	1605	1836	2631	3193	3485	3151	2723
Wind	WSAvg	6.6	6.6	6.5	6.3	5.9	5.5	5.5	5.4	6.4	7.7	7.8	7.5	7.5
Precipitation	PrecAvg	77.2	8.5	10.4	14.5	13.3	10.2	5.4	4.1	1.7	1.2	1.5	2.5	4.0
	PrecMax	101.9	15.4	17.2	25.3	21.1	20.5	9.6	8.1	6.2	3.7	4.1	6.3	10.5
	PrecMin	45.3	1.3	3.0	7.0	8.3	4.5	1.6	1.1	0.5	0.1	0.3	0.4	0.1
	PrecStd	12.7	3.8	3.7	3.8	3.4	4.2	1.9	2.0	1.2	0.9	1.1	1.4	3.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.4	91.7	91.4	89.9	89.8	90.0	91.0	91.8	93.6	95.2	94.7	93.6
		MCWB	78.3	78.9	78.5	78.5	79.0	78.0	77.4	78.2	77.4	77.3	76.8	77.5
	2%	DB	91.4	89.6	89.3	88.2	88.2	89.3	89.7	91.4	92.6	93.5	93.3	93.0
		MCWB	77.9	78.1	78.0	78.4	78.9	78.0	77.6	78.2	77.6	77.6	77.3	77.5
	5%	DB	89.6	87.9	87.7	87.6	87.7	88.0	88.2	89.9	91.5	92.8	91.8	91.5
		MCWB	77.4	78.0	78.0	78.4	78.8	78.2	78.0	77.9	77.9	77.6	77.7	77.6
	10%	DB	87.8	86.3	86.1	86.1	86.2	87.4	87.7	89.5	91.0	91.4	91.2	89.8
		MCWB	77.8	77.8	77.9	78.1	78.3	78.2	78.1	77.8	77.8	77.7	77.8	77.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.0	80.3	80.2	80.7	80.7	80.1	80.1	80.2	80.3	80.1	80.0	80.0
		MCDB	86.3	87.2	86.3	86.5	86.6	86.9	87.0	88.0	88.4	89.4	88.6	87.6
	2%	WB	79.2	79.2	79.2	79.6	79.8	79.4	79.2	79.6	79.7	79.6	79.4	79.2
		MCDB	86.3	85.9	85.3	85.3	85.7	86.3	86.4	87.7	88.1	88.8	88.3	87.4
	5%	WB	78.8	78.8	78.8	79.1	79.2	79.0	78.9	79.1	79.1	79.1	79.0	78.8
		MCDB	86.2	85.4	84.9	84.6	85.0	85.8	86.2	87.4	87.8	88.2	88.0	87.3
	10%	WB	78.2	78.2	78.3	78.6	78.8	78.5	78.4	78.7	78.7	78.6	78.6	78.3
		MCDB	85.3	84.4	84.1	84.0	84.5	85.1	85.6	87.1	87.4	87.7	87.5	86.5
Mean Daily Temperature Range	5% DB	MDBR	11.9	11.3	11.1	11.0	11.4	12.8	13.7	14.5	14.3	14.5	14.4	13.2
		MCDBR	13.6	12.8	12.4	12.2	12.6	13.9	14.6	15.1	15.0	15.9	15.3	14.3
		MCWBR	4.3	4.9	4.8	5.0	5.3	6.1	6.3	5.7	4.9	4.8	4.8	4.3
	5% WB	MCDBR	12.6	11.9	11.5	11.3	11.8	13.1	13.9	14.8	14.4	14.7	14.5	13.4
		MCWBR	4.9	4.9	5.0	5.1	5.4	6.0	6.4	6.0	5.1	5.0	5.1	4.8
Clear-Sky Solar Irradiance	taub	0.397	0.398	0.392	0.383	0.386	0.388	0.388	0.402	0.437	0.490	0.488	0.502	0.457
	taud	2.533	2.540	2.562	2.610	2.586	2.575	2.512	2.373	2.211	2.229	2.182	2.335	
	Ebn at Noon	297	297	295	291	282	277	275	271	264	267	264	278	
	Edh at Noon	35	35	34	31	31	30	33	39	47	47	49	42	
All-Sky Solar Radiation	RadAvg	1530	1485	1501	1479	1455	1556	1651	1831	1881	1820	1696	1570	
	RadStd	79	66	83	75	89	68	77	45	57	76	72	125	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page